

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4 FT	(a)	(i)		1 in 6 / $\frac{1}{6}$ / $\frac{20}{120}$		1			1	1
		(ii)		20			1			1
		(iii)		Remove those with a 6 (or any other number) facing up	1					1
		(iv)		The remaining dice were rolled again (and again) (1) taking away the 6's (or stated side) each time (1)	2					2
	(b)	(i)		19 (1) 50 (1)		2				2
		(ii)		60		1			1	1
		(iii)		4 throws			1		1	1
		(iv)		120 $\rightarrow$ 60 $\rightarrow$ 30 $\rightarrow$ 15 (1) Half-lives =3 (1) Accept 3 on answer line for 2 marks		2			2	2
	(c)	(i)		Longer <u>half-life</u> (1) So will not need replacing as often (1) OR Chromium has a shorter half-life (1) So needs replacing more often (1)			2			
		(ii)		Beta and gamma (emitter) (1) will penetrate packing (1) OR Bismuth emits alpha (1) Which won't penetrate the packing (1)			2			
				Question 4 total	3	6	6	15	5	11